The quality control department of a fabric finishing plant is studying the effect of several factors on the dyeing of cotton - synthetic cloth used to manufacture men's shirts. Two operators, three cycle times and two temperature were selected and two small specimens of the cloth were dyed under each combination of experimental conditions. The finished cloth was compared to a standard and a numerical score was assigned. The results are shown in table. Analyze the data and draw conclusions at the significance levels, of 0.05.

Temperature (B)

Cycle time (A)	250° opr - 1 (C)	250° operator 2 (C)	300° operator 1 (C)	300° operator 2 (C)	
40	23, 25	27, 26	24, 28	38, 35	226
50	36, 36	34, 39	37, 35	34, 36	257
60	28, 27	35, 34	26, 25	36, 34	245
	175	195	175	213	
	370		388		

$\text{opr-1}_{\text{total}} = 175 + 175 = 350$

$\text{Opr-2}_{\text{total}} = 195 + 213 = 408$

Given:

1) Factors:

cycle time - A

temperature - B

operator - C

2) Levels:

Interaction : 2 = n. $N = A_l * B_l * C_l * n$

$= 3 \times 2 \times 2 \times 2$

$\boxed{N = 24}$

To find:

significance level.

Soln.

$F = \frac{MS}{MS_{error}}$, $MS = \frac{SS}{df}$

$SS_{total} = SS_A + SS_B + SS_C + SS_{AB} + SS_{BC} + SS_{AC} + SS_{ABC} + SS_{error}$

Correction factor CF

$CF = \Sigma (X_{total})^2 \div N$

$= \frac{(758)^2}{24}$

$\boxed{CF = 23940.167}$

$$SSA = \frac{226^2}{8} + \frac{287^2}{8} + \frac{245^2}{8} - CF$$

$$= 24183.75 - 23940.167$$

$$\boxed{SS_A = 243.583}$$

$$SS_B = \cancel{\frac{175^2}{12} + \frac{195^2}{12} + \frac{175^2}{12} + \frac{213^2}{12} - 23940.167}$$

$$\cancel{= 24107.333 - 23940.167}$$

$$\cancel{= 167.166}$$ (Two temperatures only).

$$SS_B = \frac{370^2}{12} + \frac{388^2}{12} - 23940.167$$

$$\boxed{SS_B = 13.49}$$

$$SS_C = \frac{350^2}{12} + \frac{408^2}{12} - 23940.167$$

$$= 24080.83 - 23940.167$$

$$\boxed{SS_C = 140.166}$$

$$SS_{AB} = \frac{(23+25+27+26)^2}{4} + \frac{(24+28+38+35)^2}{4} +$$

$$\frac{(36+36+34+39)^2}{4} + \frac{(37+35+34+36)^2}{4} +$$

$$\frac{(28+27+35+34)^2}{4} + \frac{(26+25+36+34)^2}{4} - 23940.167$$

$$= \frac{(101)^2 + (125)^2 + (145)^2 + (142)^2 + (124)^2 + (121)^2}{4} - 23940.167$$

$= 24\cancel{4}258.00 - 23940.167$

$$\boxed{SS_{AB\,\text{Subtotal}} = 317.583}$$

$SS_{AB} = SS_{AB\,\text{subtotal}} - SS_A - SS_B$

$= 317.583 - 243.584 - 13.49$

$$\boxed{SS_{AB} = \cancel{214.509}\ 60.759}$$

$SS_{BC} \rightarrow$ Temperature & operator.

$$SS_{BC\,\text{subtotal}} = \frac{175^2}{6} + \frac{195^2}{6} + \frac{175^2}{6} + \frac{213^2}{6} - 23940.167$$

$= 24107.33 - 23940.167$

$SS_{BC\,\text{subtotal}} = 167.66$

$SS_{BC} = SS_{BC\,(\text{subtotal})} - SS_B - SS_C$

$= 167.66 - 13.49 - 140.166$

$$\boxed{SS_{BC} = 13.51}$$

$$SS_{AC\,(\text{Subtotal})} = \frac{(28+25+24+28)^2}{4} + \frac{(27+26+38+35)^2}{4} + \frac{(36+37+36+35)^2}{4}$$

$$+ \frac{(34+34+39+36)^2}{4} + \frac{(28+27+26+25)^2}{4} + \frac{(35+34+36+34)^2}{4} - CF$$

$$= \frac{100^2}{4} + \frac{126^2}{4} + \frac{144^2}{4} + \frac{143^2}{4} + \frac{106^2}{4} + \frac{139^2}{4} - 23940.167$$

$= 24404.4 - 23940.167$

$$\boxed{SS_{AC\,St} = 464.33}$$

$$SS_{AC} = \underset{(\text{subtotal})}{SS_{AC}} - SS_A - SS_C$$

$$= 464.33 - 243.584 - 140.166$$

$$\boxed{SS_{AC} = 80.584}$$

$$SS_{total} = \Sigma x^2 - CF$$

$$= 24550 - 23940.167$$

$$\boxed{SS_{total} = 609.833}$$

$$SS_{error} = SS_{total} - (SS_A + SS_B + SS_C + SS_{AB} + SS_{BC} + SS_{ABC})$$

$$\underset{\text{subtotal}}{SS_{ABC}} = \frac{(23+25)^2}{2} + \frac{(27+26)^2}{2} + \frac{(24+28)^2}{2} + \frac{(38+35)^2}{2} +$$

$$\frac{(36+36)^2}{2} + \frac{(54+39)^2}{2} + \frac{(37+35)^2}{2} + \frac{(34+36)^2}{2}$$

$$+ \frac{(28+27)^2}{2} + \frac{(35+34)^2}{2} + \frac{(26+25)^2}{2} + \frac{(36+34)^2}{2} - CF$$

$$= (48^2 + 53^2 + 52^2 + 73^2 + 72^2 + 93^2 + 72^2 + 70^2 +$$
$$55^2 + 69^2 + 51^2 + 70^2)/2 - 23490.167$$

$$= 24515 - 23490.167$$

$$\boxed{\underset{(st)}{SS_{ABC}} = 574.833.}$$

$$SS_{ABC} = \underset{(st)}{SS_{ABC}} - SS_A - SS_B - SS_C - SS_{AB} - SS_{BC} - SS_{AC}$$

$$= 574.83 - (243.584 + 13.49 + 140.166 + 60.759 + 13.51 + 80.584)$$

$$\boxed{SS_{ABC} = 22.737}$$

$SS_{err} = 609.833 - 60.759 - 13.51 - 80.584 - 22.737 - 243.58$
$- 140.166 - 13.5.$

$\boxed{SS_{err} = 34.73}$

<u>Calculate degree of freedom:</u>

$df_A = A-1 = 3-1 = 2$

$df_B = B-1 = 2-1 = 1$

$df_C = C-1 = 2-1 = 1$

$df_{AB} = (A-1)(B-1) = 2$

$df_{BC} = (B-1)(C-1) = 1$

$df_{AC} = (A-1)(C-1) = 2$

$(A-1)(B-1)(C-1) = 2 \times 1 \times 1 = 2.$

$df_{ABC} = \cancel{(A-1)(B-1)(C-1) = 2 \times 1 \times 2 = 4}$

$\cancel{ABC(A-1) = 2 \times 1 \times 1 \times (2-1)}$

$\cancel{df_{ABC} = 2}$

$df_{error} = df_{total} - \text{sum of trails}$

$= (N-1) - (2+1+1+2+1+2+2)$

$= 23 - 11$

$df_{error} = 12.$

Mean square (MS) $MS = \frac{SS}{df}$

$MS_A = \frac{243.583}{2} = 121.7915$

$MS_B = \frac{13.49}{1} = 13.49$

$MS_C = \frac{140.166}{1} = 140.166$

$MS_{AB} = \frac{60.759}{2} = 30.379$

$MS_{BC} = \frac{13.51}{1} = 13.51$

$MS_{AC} = \frac{80.584}{2} = 40.292$

$MS_{ABC} = \frac{22.737}{2} = 11.3685$

$MS_{error} = \frac{34.73}{12} = 2.894$

F for A, B, C, AB, AC, BC, ABC

$F_A = \frac{MSA}{MSer} = \frac{121.7915}{2.894} = 42.084$

$F_B = \frac{MSB}{MSer} = \frac{13.49}{2.894} = 4.6613$

$F_C = \frac{MSc}{MSerr} = \frac{140.166}{2.894} = 48.4333$

$F_{AB} = \frac{MSAB}{MSerr} = \frac{30.379}{2.894} = 10.4972$

$F_{BC} = \frac{13.51}{2.894} = 4.6682$

$F_{AC} = \frac{40.292}{2.894} = 13.9225$

$F_{ABC} = \frac{11.3685}{2.894} = 3.9282.$

ANOVA table

source of variation	df	SS	MS	F	Factual 5%	
A	2	243.583	121.791	42.084 >	3.88	sig
B	1	13.49	13.49	4.6613 <	4.75	not sig
C	1	140.166	30.379	48.43 >	4.75	sig
AB	2	60.759	140.166	10.4972 >	3.88	sig
BC	1	13.51	13.51	4.6682 <	4.75	not sig
AC	2	80.584	40.292	13.9225 >	3.88	sig
ABC	2	22.737	11.3685	3.9282. >	3.88	sig.
exp. err.	12	34.73.	2.894			
Total	23	609.833.				

Note : sig - significant.

07.11.2025.

Taguchi Method

Taguchi defines quality loss as :

"the loss imparted to soceity from the time a product is shipped".

Quality must be "engineered in"

Contributions of Taguchi

1) Quality engineering philosophy : Target and loss function
2) Methodology : system, parameter, tolerance design
3) Experimental design : Use of orthogonal arrays
4) Analysis : Use signal to Noise ratio (S/N ratio)

N → external disturbance (or) factors

S → output.

Eg : characteristics such as fuel economy of a car,
the weight of a package of breakfast cereal,
the power losses of a home hot water heater.

Competitive → Robust

The product performance must be immune to the noise variable.

Robust Design and Quality in the Product Development Process

Planning → Concept Development → System level design → Detail Design → Testing and Refinement → Product Ramp up

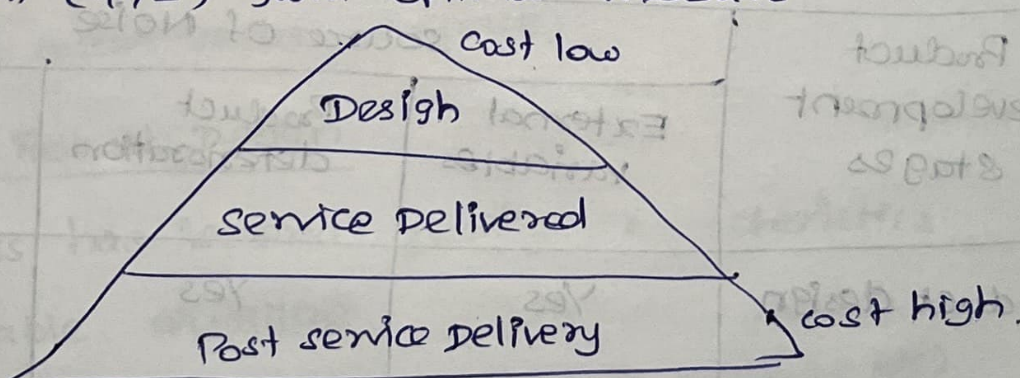
↑ (Concept Development, System level design): Robust control and system Design

↑ (Detail Design): Robust Parameter & Tolerance Design

↑ (Testing and Refinement, Product Ramp up): Quality efforts for

Basic Ideas :

* customer satisfaction
* Design to the highest standards early in the process to eliminate all non-random errors
* Quality Loss = Loss to society quantified through "Quality Loss Function".
* variation (+/−) from optimal measure



Quality Loss concept

* Deviation from target results in loss
  - → Lower than target
  - → greater than target.

Classifications of factors

1) control factors : Design factors that are to be set at optimal levels to improve quality

2) Noise factors

3) Adjustment factors :

Noise

→ 1. External Noise

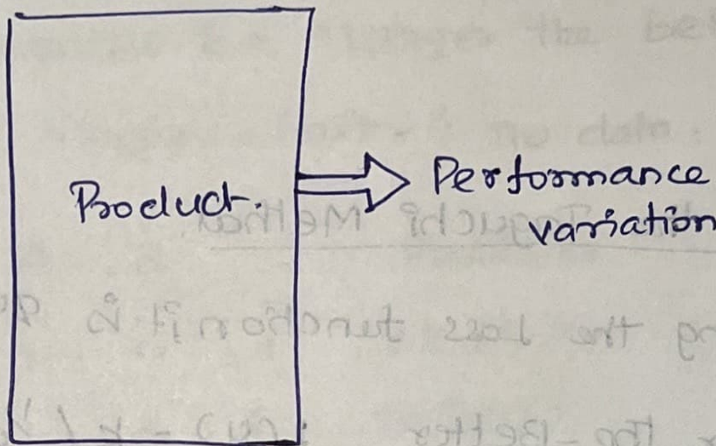
→ 2. Deterioration Noise

→ 3. Internal Noise

Product development stages

Product Development stages	Source of Noise		
	External variables	Product deterioration	Manufacturing variation
Product design	Yes	Yes	Yes
Process design	NO	NO	Yes
Manufacturing	NO	NO	Yes

A Product's Life



Quality Loss Function:

$$L(Y) = k(Y - m)^2$$

where

$L(Y)$ = Loss

k = constant = cost to correct / customer tolerance

Y = reported value

m = mean value (average)

Taguchi Robust Design

* A process has "several" quality characteristics

→ Desirable Qualities

- Larger-the-Better (strength, mileage etc)
- Nominal-the-Better (specified dimensions, uniformity)

→ Undesirable Properties

- smaller the better (defects, resources - material / time, cost)

* Determine the best setting such that

→

→.

Exploring the Taguchi Method.

* Considering the Loss function it is quantifiable

→ Larger- the -Better $L(y) = k(1/y^2)$

→ Smaller the better $L(y) = ky^2$

→ Nominal the better $L(y) = k(y-m)^2$

where m → target of the process specification.

Quality Evaluations (S/N Ratio)

control factors

A B C

input → Product → output

$$S/N = \pm 10 \log \left(\frac{\mu^2}{\sigma^2}\right)$$

+ → larger better

− → smaller better

μ = mean

σ = standard Deviation (or variation)

Note:

Taguchi follows partial factorial Design

1 → Low
2 → High } 2 level

1 - Low
2 - medium
3 → High. } 3 level

Problem

1. Three control factors A, B, C each at 3 levels are studied using an L9 orthogonal array. Each run is repeated 3 times. The measured response is a "larger the better" quality characteristics (higher = better). The data:

Run	A	B	C	Replicates
1	1	1	1	65, 67, 66
2	1	2	2	70, 72, 71
3	1	3	3	75, 74, 76
4	2	1	2	68, 69, 70
5	2	2	3	73, 75, 74
6	2	3	1	64, 63, 65
7	3	1	3	78, 77, 79
8	3	2	1	66, 68, 67
9	3	3	2	71, 70, 72.

Trail : 8

$$\frac{S}{N} = -10 \log_{10} \left[\frac{1}{n} \sum_{i=1}^{n} \frac{1}{y_i^2} \right]$$

$n = 3$

$i = 8.$

$$\frac{S}{N} = -10 \log_{10} \left[\frac{1}{3} \left(\frac{1}{y_8^2} \right) \right]$$

$$= -10 \log_{10} \left[\frac{1}{3} \left(\frac{1}{66^2} + \frac{1}{68^2} + \frac{1}{67^2} \right) \right]$$

$$= -10 \log_{10} (2.2286 \times 10^{-4}) = 36.5196.$$

$n=3$
Run $=9$

$$\frac{S}{N} = -10\log_{10}\left(\frac{1}{3}\left[\frac{1}{71^2}+\frac{1}{70^2}+\frac{1}{72^2}\right]\right)$$

$$= -10\log_{10}(1.98452\times10^{-4})$$

$$= 37.0284.$$

$n=3$
Run $=1$.

$$\frac{S}{N} = -10\log_{10}\left(\frac{1}{3}\left[\frac{1}{65^2}+\frac{1}{67^2}+\frac{1}{66^2}\right]\right)$$

$$= -10\log_{10}(2.29679\times10^{-4})$$

$$= 36.3889$$

$n=3$
Run $=2$.

$$\frac{S}{N} = -10\log\left(\frac{1}{3}\left[\frac{1}{70^2}+\frac{1}{72^2}+\frac{1}{71^2}\right]\right)$$

$$= -10\log_{10}(1.98452\times10^{-4})$$

$$= 37.0234$$

$n=3$
Run $=3$.

$$\frac{S}{N} = -10\log_{10}\left(\frac{1}{3}\left[\frac{1}{75^2}+\frac{1}{74^2}+\frac{1}{76^2}\right]\right)$$

$$= -10\log_{10}(1.77841\times10^{-4})$$

$$= 37.4996$$

$n=3$

Run $=4$

$$\frac{S}{N} = -10\log_{10}\left[\frac{1}{3}\left(\frac{1}{68^2}+\frac{1}{69^2}+\frac{1}{70^2}\right)\right]$$

$$= -10\log_{10}(2.10128\times10^{-4})$$

$$= 36.7752.$$

$n=3$

Run $=5$

$$\frac{S}{N} = -10\log_{10}\left[\frac{1}{3}\left(\frac{1}{78^2}+\frac{1}{75^2}+\frac{1}{74^2}\right)\right]$$

$$= -10\log_{10}(1.8268\times10^{-4})$$

$$= 37.3830$$

$n=3$

Run $=6$

$$\frac{S}{N} = -10\log\left[\frac{1}{3}\left(\frac{1}{64^2}+\frac{1}{63^2}+\frac{1}{65^2}\right)\right]$$

$$= -10\log_{10}(2.4425\times10^{-4})$$

$$= 36.1215$$

$n=3$

Run $=7$

$$\frac{S}{N} = -10\log_{10}\left[\frac{1}{3}\left(\frac{1}{78^2}+\frac{1}{77^2}+\frac{1}{79^2}\right)\right]$$

$$= -10\log_{10}(1.6441\times10^{-4})$$

$$= 37.8405$$

$$\text{Grand mean} = \frac{36.3889+37.0234+37.4997+36.7752+37.3830+36.1215+37.8405+36.5196+37.0234}{9}$$

$$\boxed{\text{Grand mean} = 36.9528.}$$

2) Calculate Average by factor level.

Factor A (1,2,3)

A_1 mean $\frac{S}{N}$ ratio. ⇒

$$\frac{S}{N}\text{ ratio} = -10\log_{10}\left(\frac{1}{3}\left[\frac{1}{65^2}+\frac{1}{70^2}+\frac{1}{75^2}\right]\right)$$

$$= -10\log_{10}(2.06181\times 10^{-4})$$

$$= 36.8575$$

A_2 mean $\frac{S}{N}$ ratio ⇒

$$\frac{S}{N}\text{ ratio} = -10\log_{10}\left[\frac{1}{3}\left(\frac{1}{68^2}+\frac{1}{78^2}+\frac{1}{74^2}\right)\right]$$

$$= -10\log(2.16018\times 10^{-4})$$

$$= 36.6550.$$

A_3 mean $\frac{S}{N}$ ratio ⇒

$$\frac{S}{N}\text{ ratio} = -10\log_{10}\left[\frac{1}{3}\left(\frac{1}{78^2}+\frac{1}{66^2}+\frac{1}{71^2}\right)\right]$$

$$= -10\log_{10}(1.97435\times 10^{-4})$$

$$= 37.0457$$

Factor B (1,2,3)

B1 mean $\frac{S}{N}$ ratio ⇒

$$\frac{S}{N} \text{ ratio} = -10 \log_{10} \left(\frac{1}{3} \left[\frac{1}{67^2} + \frac{1}{69^2} + \frac{1}{77^2} \right] \right)$$

$$= -10 \log_{10} (2.0048 \times 10^{-4})$$

$$= 36.9790.$$

B_2 mean $\frac{S}{N}$ ratio

$$S/N \text{ ratio} = -10 \log_{10} \left(\frac{1}{3} \left[\frac{1}{72^2} + \frac{1}{75^2} + \frac{1}{68^2} \right] \right)$$

$$= -10 \log_{10} (1.9564 \times 10^{-4})$$

$$= 37.0852$$

B3 mean $\frac{S}{N}$ ratio.

$$\frac{S}{N} \text{ ratio} = -10 \log_{10} \left[\frac{1}{3} \left(\frac{1}{74^2} + \frac{1}{63^2} + \frac{1}{70^2} \right) \right]$$

$$= -10 \log_{10} (2.1288 \times 10^{-4})$$

$$= 36.7185.$$

Factor C (1,2,3)

C1 mean $\frac{S}{N}$ ratio ⇒

$$\frac{S}{N} \text{ ratio} = -10 \log_{10} \left(\frac{1}{3} \left[\frac{1}{66^2} + \frac{1}{65^2} + \frac{1}{67^2} \right] \right)$$

$$= -10 \log (2.2967 \times 10^{-4})$$

$$= 36.3888$$

C2 mean S/N ratio ⇒

$$\frac{S}{N} \text{ ratio} = -10\log_{10}\left[\frac{1}{3}\left(\frac{1}{71^2}+\frac{1}{70^2}+\frac{1}{72^2}\right)\right]$$

$$= -10\log_{10}(1.9845\times10^{-4})$$

$$= 37.0234.$$

C3 mean $\frac{S}{N}$ ratio ⇒

$$S/N \text{ ratio} = -10\log_{10}\left[\frac{1}{3}\left(\frac{1}{76^2}+\frac{1}{74^2}+\frac{1}{79^2}\right)\right]$$

$$= -10\log_{10}(1.7199\times10^{-4})$$

$$= 37.6449.$$

Calculate correction factor CF

$$CF = \frac{(\Sigma X_{total})^2}{N} \qquad N = 9. \qquad CF = \frac{\left[\sum_{i=1}^{9}(S/N)_i\right]^2}{N}$$

$$\cancel{\Sigma X_A = 333.0752}$$

$$\cancel{\Sigma X_B = 332.5752}$$

$$\cancel{\Sigma X_C = 333.0753}$$

$$\cancel{CF = \frac{(333.0752+332.5752+333.0753)^2}{9}}$$

$$\cancel{CF = 110828.1138}$$

$$= \frac{(332.5752)^2}{9}$$

$$\boxed{CF = 12289.5848}$$

	A	B	C
1	36.85	36.97	36.38
2	36.65	37.08	37.02
3	37.04	36.71	37.64.

		A	B	C
S/N ratio.	1	36.3889	36.3889	36.3889
		37.0234	36.7752	36.1215
		37.4996	37.8405	36.5196
	2	36.7752	37.0234	37.0234
		37.3830	37.3830	36.7752
		36.1215	36.5196	37.0234
	3	37.8405	37.4997	37.4997
		36.5196	36.1215	37.3830
		37.0234	37.0234	37.8405.

SS → sum of squares.

$$SS_A = \frac{\Sigma A_1^2}{n} + \frac{\Sigma A_2^2}{n} + \frac{\Sigma A_3^2}{n} - CF.$$

$$= \frac{110.9120^2}{3} + \frac{110.2797^2}{3} + \frac{111.3835^2}{3} - 12289.5848$$

$$= 12289.789 - 12289.5848$$

$$\boxed{SS_A = 0.2042}$$

$$SS_B = \frac{111.0046^2}{3} + \frac{110.9260^2}{3} + \frac{110.6446^2}{3} - CF$$

$$= 12,289.6087 - 12289.5848$$

$$\boxed{SS_B = 0.02393}$$

$$SS_C = \frac{109.03^2}{3} + \frac{110.8220^2}{3} + \frac{112.7232^2}{3} - CF$$

$$= 12291.8588 - 12289.5848$$

$$\boxed{SS_C = 2.2740}$$

$$SS_{total} = \sum_{i=1}^{9} (S/N)_i^2 - CF$$

$$= (36.3889^2 + 37.0234^2 + 37.4997^2 + 36.7752^2 + 37.3830^2 + 36.1215^2 + 37.8405^2 + 36.5196^2 + 37.0234^2) - 12289.5848$$

$$= 12292.0952 - 12289.5848$$

$$\boxed{SS_{total} = 2.5104}$$

$$SS_{error} = SS_{total} - SS_A - SS_B - SS_C$$

$$= 2.5104 - 0.2042 - 0.02393 - 2.2740$$

$$\boxed{SS_{error} = 0.00827.}$$

Calculate Degree of freedom (df)

$df_A = A-1 = (3-1) = 2$

$df_B = B-1 = 3-1 = 2$

$df_C = C-1 = 3-1 = 2$

$df_{error} = df_{total} - [(A-1)+(B-1)+(C-1)] = 8 - (2+2+2) = 2$

$df_{total} = N-1 = 9-1 = 8$

Calculate Mean square (MS)

$$MS_A = \frac{SS_A}{df_A} = \frac{0.2042}{2} = 0.1021$$

$$MS_B = \frac{SS_B}{df_B} = \frac{0.02393}{2} = 0.011965$$

$$MS_C = \frac{SS_C}{df_C} = \frac{2.2740}{2} = 1.1370$$

$$MS_{err} = \frac{SS_{err}}{df_{err}} = \frac{0.00827}{2} = 0.004135.$$

Calculate F_{factor} for A, B, C.

$$F_A = \frac{MS_A}{MS_{err}} = \frac{0.1021}{0.004135} = 24.691$$

$$F_B = \frac{MS_B}{MS_{error}} = \frac{0.011965}{0.004135} = 2.893$$

$$F_C = \frac{MS_C}{MS_{err.}} = \frac{1.137}{0.004135} = 274.96$$

ANOVA Table

Source of variation	SS	df	MS	F_{factor}	% Contribution
A	0.2042	2	0.1021	24.691	8.134 %
B	0.02393	2	0.011965	2.893	0.95 %
C	2.2740	2	1.1370	274.96	90.58 %
Exp. Error	0.00827	2	0.004135		
Total	2.5104	8			

$$\% \text{ contribution} = \frac{SS_{factor}}{SS_{Total}} \times 100\ \%$$

$$\% \text{ Contribution}_A = \frac{SS_A}{SS_{total}} \times 100$$

$$= \frac{0.2042}{2.5104} \times 100 = 8.134\ \%$$

$$=$$

$$\% \text{ Contribution}_B = \frac{SS_B}{SS_{total}} \times 100$$

$$= \frac{0.02393}{2.5104} = 0.95\ \%$$

$$\% \text{ Contribution}_C = \frac{SS_C}{SS_{total}} = \frac{2.2740}{2.5104} = 90.58\ \%$$

Inference

(i) Factor C has the highest contribution (90.5%) indicating it is the most significant factor influencing S/N ratio

(ii) Factor B shows negligible influence (0.95%) (ie) it has small effect.

Predicted S/N = grand mean + sum of levels chosen

= 36.9528 + 111.6956

=

$$y = 10^{(\text{Pred } S/N / 20)}$$

$$= 10^{(37.8/20)}$$

42.2041

42.3241

42.5873

42.3869

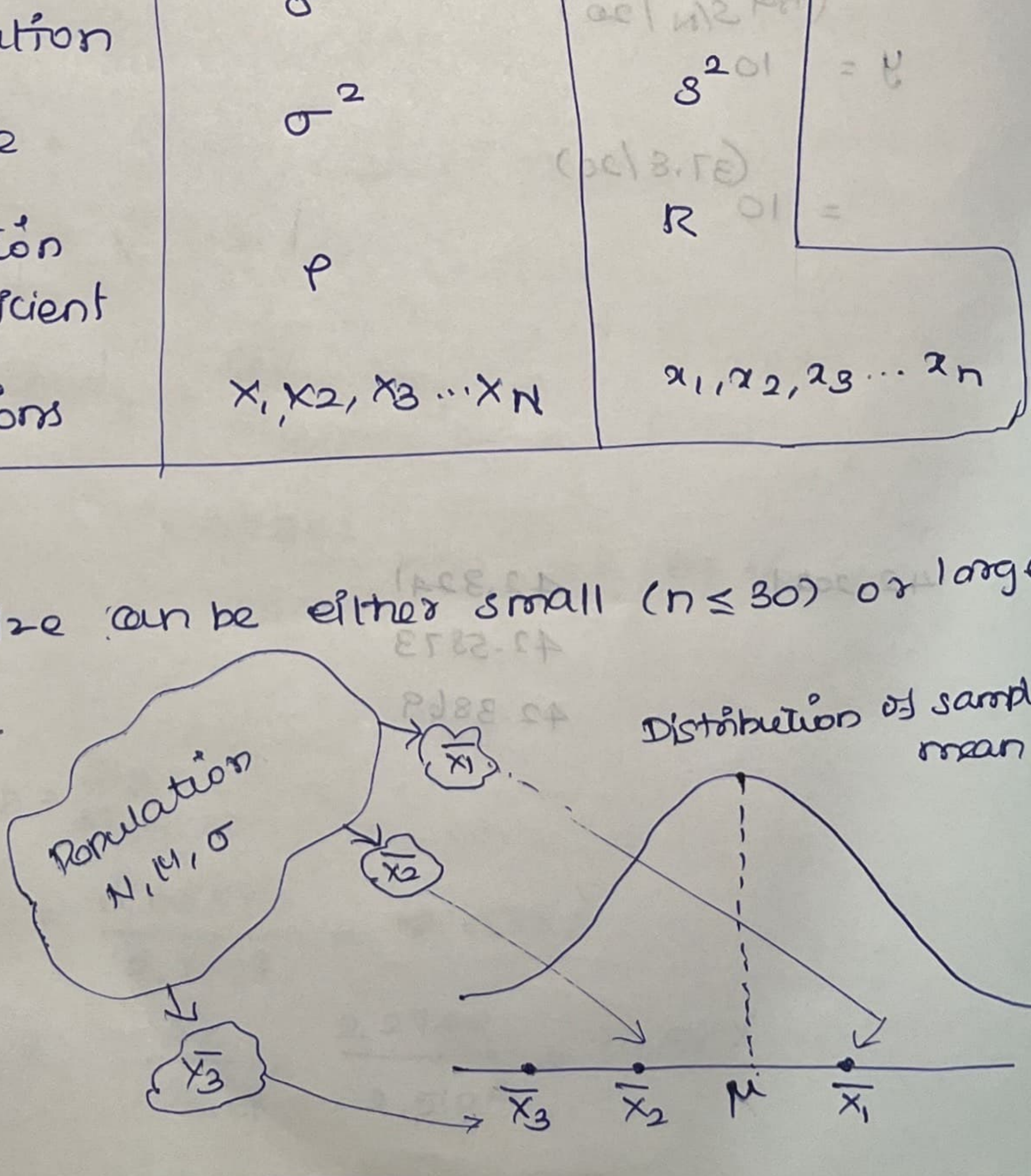
Sampling statistics and sizes

* A "sample" is a carefully chosen representative part of the "population" use to draw inference about the population.

* Parameter is a number that describe the property of entire population, whereas "statistic" is a no. that describe the Properties of a sample.

	Population	sample
size	N	n
Mean	μ	$\bar{x}$
Standard Deviation	σ	S
Variance	σ^2	S^2
correlation coefficient	ρ	R
observations	$X_1, X_2, X_3 \dots X_N$	$x_1, x_2, x_3 \dots x_n$

* Sample size can be either small ($n \leq 30$) or large ($n > 30$).



Sampling Distributions

The sampling distributions for small sample sizes are

1. Student's t-distribution ($n \leq 30$)

2. Chi square (χ^2) Distribution (Both $n \leq 30$ and $n > 30$)

3. F-Distribution ($n \leq 30$).

1. Student's T Distribution

If $x_1, x_2, x_3 \ldots x_n$ is a sample of size "n" from a normal population with unknown μ and unknown variance σ^2 and if

$$\bar{x} = \frac{\Sigma x_i}{n} \text{ and } s^2 = \frac{\Sigma (x_i - \bar{x})^2}{n-1}$$ Then the

$$\text{statistic } t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1} \text{ df}.$$

$\sim \rightarrow$ follows and df means degree of freedom.

The t-test used to compare

* one sample t-test for sample mean and population mean
* Paired t-test for sample Means before and after any change inducing event, say a treatment of population
* Two sample t-Test for means of two different samples
* Difference b/w sample correlation coefficient and population correlation coefficient.

1) one sample T-test Eg:

The population mean and sample mean are compared.

$$t\text{-test} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Eg1: A retail store wants to improve sales. Historical sales data indicate that the average sale was Rs100 per transaction. After training the sale force recent sales Data (taken from a sample of 25 salespeople) indicates an average scale of Rs 130 with a standard deviation of Rs 15. Did the training work? Assume that the level of significance is 5%. Also assume that the t-critical value is 1.71.